

## WAUKEGAN HARBOR, IL, USA

WMO: 997735

Lat: 42.361N

Lon: 87.813W

Elev: 617

StdP: 14.37

Time Zone: -6.00 (NAC)

Period: 06-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB  |            | Humidification DP/MCDB and HR |            |            |            |            |            | Coldest Month WS/MCDB |             |             |             | MCWS/PCWD to 99.6% DB |            | WSF        |
|---------------|-------------|------------|-------------------------------|------------|------------|------------|------------|------------|-----------------------|-------------|-------------|-------------|-----------------------|------------|------------|
|               |             |            | 99.6%                         |            |            | 99%        |            |            | 0.4%                  |             | 1%          |             |                       |            |            |
|               | 99.6%       | 99%        | DP                            | HR         | MCDB       | DP         | HR         | MCDB       | WS                    | MCDB        | WS          | MCDB        | MCWS                  | PCWD       |            |
| (a)<br>1      | (b)<br>-1.7 | (c)<br>3.0 | (d)<br>N/A                    | (e)<br>N/A | (f)<br>N/A | (g)<br>N/A | (h)<br>N/A | (i)<br>N/A | (j)<br>31.5           | (k)<br>31.3 | (l)<br>28.7 | (m)<br>29.4 | (n)<br>14.3           | (o)<br>270 | (p)<br>N/A |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |            |             |            |             |            | Evaporation WB/MCDB |            |            |            |            |            | MCWS/PCWD to 0.4% DB |            |
|---------------|------------------------|-----------------|------------|-------------|------------|-------------|------------|---------------------|------------|------------|------------|------------|------------|----------------------|------------|
|               |                        | 0.4%            |            | 1%          |            | 2%          |            | 0.4%                |            | 1%         |            | 2%         |            |                      |            |
|               |                        | DB              | MCWB       | DB          | MCWB       | DB          | MCWB       | WB                  | MCDB       | WB         | MCDB       | WB         | MCDB       | MCWS                 | PCWD       |
| (a)<br>7      | (b)<br>11.6            | (c)<br>86.7     | (d)<br>N/A | (e)<br>83.6 | (f)<br>N/A | (g)<br>80.4 | (h)<br>N/A | (i)<br>N/A          | (j)<br>N/A | (k)<br>N/A | (l)<br>N/A | (m)<br>N/A | (n)<br>N/A | (o)<br>14.7          | (p)<br>220 |

| Dehumidification DP/MCDB and HR |            |            |            |            |            |            |            |            | Enthalpy/MCDB |            |            |            |            |            | Extreme<br>Max WB |
|---------------------------------|------------|------------|------------|------------|------------|------------|------------|------------|---------------|------------|------------|------------|------------|------------|-------------------|
| 0.4%                            |            |            | 1%         |            |            | 2%         |            |            | 0.4%          |            | 1%         |            | 2%         |            |                   |
| DP                              | HR         | MCDB       | DP         | HR         | MCDB       | DP         | HR         | MCDB       | Enth          | MCDB       | Enth       | MCDB       | Enth       | MCDB       |                   |
| (a)<br>N/A                      | (b)<br>N/A | (c)<br>N/A | (d)<br>N/A | (e)<br>N/A | (f)<br>N/A | (g)<br>N/A | (h)<br>N/A | (i)<br>N/A | (j)<br>N/A    | (k)<br>N/A | (l)<br>N/A | (m)<br>N/A | (n)<br>N/A | (o)<br>N/A | (p)<br>N/A        |

## Extreme Annual Design Conditions

| Extreme Annual WS |          |          | Extreme Annual Temperature |          |                    |         | n-Year Return Period Values of Extreme Temperature |          |            |          |            |          |            |           |  |  |
|-------------------|----------|----------|----------------------------|----------|--------------------|---------|----------------------------------------------------|----------|------------|----------|------------|----------|------------|-----------|--|--|
|                   |          |          | Mean                       |          | Standard Deviation |         | n=5 years                                          |          | n=10 years |          | n=20 years |          | n=50 years |           |  |  |
| 1%                | 2.5%     | 5%       | Min                        | Max      | Min                | Max     | Min                                                | Max      | Min        | Max      | Min        | Max      | Min        | Max       |  |  |
| (a) 30.9          | (b) 26.9 | (c) 24.4 | (e) -6.8                   | (f) 91.6 | (g) 6.4            | (h) 4.3 | (i) -11.4                                          | (j) 94.6 | (k) -15.1  | (l) 97.2 | (m) -18.7  | (n) 99.6 | (o) -23.3  | (p) 102.7 |  |  |
|                   |          |          | DB                         |          |                    |         |                                                    |          |            |          |            |          |            |           |  |  |
|                   |          |          | WB                         |          |                    |         |                                                    |          |            |          |            |          |            |           |  |  |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 48.5   | 23.2  | 24.4  | 35.4  | 45.3  | 55.2  | 65.6  | 71.6  | 70.8  | 64.3  | 53.3  | 41.6  | 30.2  |  |  |
|                                                                        | DBStd       | 18.83  | 11.65 | 10.36 | 8.94  | 7.24  | 7.73  | 7.13  | 6.15  | 5.08  | 6.83  | 7.39  | 9.32  | 9.91  |  |  |
|                                                                        | HDD50       | 3171   | 832   | 717   | 464   | 181   | 33    | 0     | 0     | 0     | 1     | 52    | 276   | 615   |  |  |
|                                                                        | HDD65       | 6611   | 1297  | 1137  | 919   | 593   | 318   | 80    | 15    | 13    | 92    | 366   | 702   | 1079  |  |  |
|                                                                        | CDD50       | 2637   | 0     | 0     | 10    | 39    | 194   | 467   | 671   | 646   | 431   | 154   | 24    | 1     |  |  |
|                                                                        | CDD65       | 601    | 0     | 0     | 0     | 1     | 14    | 97    | 220   | 193   | 72    | 4     | 0     | 0     |  |  |
|                                                                        | CDH74       | 3102   | 0     | 0     | 1     | 11    | 109   | 560   | 1340  | 787   | 282   | 12    | 0     | 0     |  |  |
|                                                                        | CDH80       | 758    | 0     | 0     | 0     | 0     | 19    | 144   | 390   | 147   | 57    | 1     | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 11.7   | 13.5  | 13.1  | 12.8  | 13.4  | 11.6  | 9.9   | 9.1   | 9.1   | 10.5  | 12.0  | 13.2  | 12.8  |  |  |
| (15) Precipitation                                                     | PrecAvg     | 35.9   | 1.9   | 1.6   | 2.2   | 3.8   | 4.0   | 4.1   | 3.5   | 3.9   | 3.2   | 3.0   | 2.7   | 2.0   |  |  |
|                                                                        | PrecMax     | 44.5   | 3.7   | 4.6   | 4.4   | 7.6   | 8.4   | 9.7   | 7.2   | 11.3  | 9.5   | 6.9   | 5.9   | 4.5   |  |  |
|                                                                        | PrecMin     | 23.6   | 0.5   | 0.2   | 0.7   | 1.7   | 0.6   | 1.0   | 1.3   | 1.9   | 0.7   | 0.7   | 0.4   | 0.8   |  |  |
|                                                                        | PrecStd     | 4.8    | 0.9   | 1.0   | 1.1   | 1.6   | 1.8   | 2.2   | 1.6   | 1.9   | 1.7   | 1.8   | 1.5   | 1.0   |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 55.1  | 55.4  | 67.2  | 75.3  | 82.8  | 88.8  | 93.8  | 88.2  | 86.3  | 75.3  | 65.4  | 56.7  |  |  |
|                                                                        |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |  |  |
|                                                                        | 2%          | DB     | 47.9  | 46.7  | 59.4  | 68.6  | 77.3  | 84.4  | 87.5  | 84.6  | 80.5  | 70.5  | 61.0  | 49.7  |  |  |
|                                                                        |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |  |  |
|                                                                        | 5%          | DB     | 42.5  | 40.8  | 52.7  | 62.0  | 72.9  | 80.4  | 84.2  | 80.5  | 77.0  | 66.8  | 57.0  | 46.0  |  |  |
|                                                                        |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |  |  |
|                                                                        | 10%         | DB     | 38.3  | 36.5  | 47.0  | 55.8  | 68.3  | 77.0  | 80.9  | 78.3  | 74.0  | 64.0  | 53.6  | 42.4  |  |  |
|                                                                        |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |  |  |
|                                                                        |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |  |  |
|                                                                        | 2%          | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |  |  |
|                                                                        |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |  |  |
|                                                                        | 5%          | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |  |  |
|                                                                        |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |  |  |
|                                                                        | 10%         | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |  |  |
|                                                                        |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 12.7  | 12.0  | 10.8  | 11.7  | 13.3  | 12.5  | 11.6  | 10.7  | 11.5  | 11.8  | 11.5  | 11.1  |  |  |
|                                                                        |             | MCDBR  | 16.3  | 15.4  | 20.5  | 22.8  | 22.1  | 19.2  | 16.8  | 14.4  | 15.6  | 15.9  | 15.4  | 13.2  |  |  |
|                                                                        |             | MCWBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |  |  |
|                                                                        | 5% WB       | MCDBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |  |  |
|                                                                        |             | MCWBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.294 | 0.309 | 0.343 | 0.397 | 0.423 | 0.439 | 0.439 | 0.433 | 0.404 | 0.353 | 0.313 | 0.294 |  |  |
|                                                                        | taud        |        | 2.453 | 2.406 | 2.399 | 2.276 | 2.257 | 2.262 | 2.260 | 2.277 | 2.329 | 2.471 | 2.523 | 2.490 |  |  |
|                                                                        | Ebn at Noon |        | 271   | 284   | 286   | 276   | 271   | 266   | 264   | 262   | 262   | 265   | 262   | 261   |  |  |
|                                                                        | Edh at Noon |        | 24    | 29    | 33    | 41    | 43    | 43    | 42    | 40    | 36    | 27    | 22    | 21    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 471   | 749   | 1124  | 1459  | 1673  | 1918  | 1943  | 1684  | 1372  | 891   | 557   | 391   |  |  |
|                                                                        | RadStd      |        | 59    | 91    | 92    | 120   | 132   | 127   | 89    | 102   | 117   | 100   | 56    | 44    |  |  |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       | Degree-Days |       |       |       |       |      |
|------------------------------|--|-------|---------|--------|---------|-------|-------------|-------|-------|-------|-------|------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP       | HDD50 | HDD65 | CDD50 | CDD65 |      |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)         | (j)   | (k)   | (l)   | (m)   |      |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A         | N/A   | N/A   | N/A   | N/A   | (46) |
| (47) Regional (57 neighbors) |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A         | N/A   | N/A   | N/A   | N/A   | (47) |

Nomenclature: See separate page